



Problem Challenge 2

We'll cover the following



- Structurally Unique Binary Search Trees (hard)
- Try it yourself

Structurally Unique Binary Search Trees (hard)

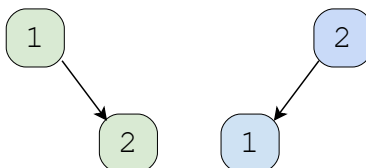
Given a number 'n', write a function to return all structurally unique Binary Search Trees (BST) that can store values 1 to 'n'?

Example 1:

Input: 2

Output: List containing root nodes of all structurally unique BSTs.

Explanation: Here are the 2 structurally unique BSTs storing all numbers from 1 to 2:

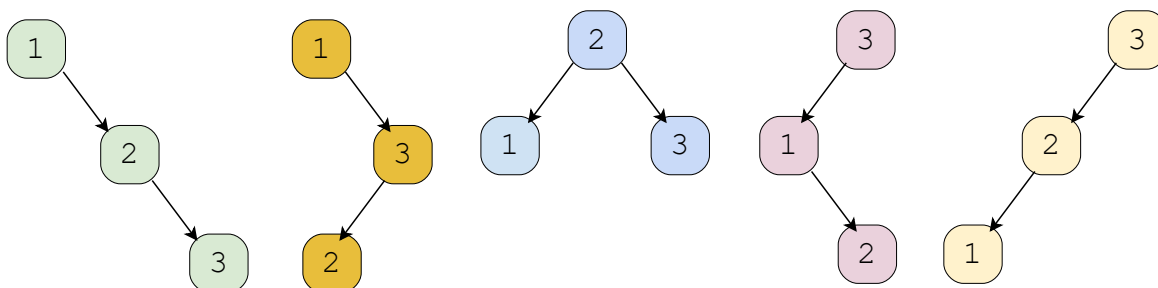


Example 2:

Input: 3

Output: List containing root nodes of all structurally unique BSTs.

Explanation: Here are the 5 structurally unique BSTs storing all numbers from 1 to 3:



Try it yourself

Try solving this question here:



Java

Python3

JS

C++

```
1 import java.util.*;
2
3 class TreeNode {
4     int val;
5     TreeNode left;
6     TreeNode right;
7
8     TreeNode(int x) {
9         val = x;
10    }
11 };
12
13 class UniqueTrees {
14     public static List<TreeNode> findUniqueTrees(int n) {
15         List<TreeNode> result = new ArrayList<>();
16         // TODO: Write your code here
17         return result;
18     }
19
20     public static void main(String[] args) {
21         List<TreeNode> result = UniqueTrees.findUniqueTrees(2);
22         System.out.print("Total trees: " + result.size());
23     }
24 }
25
```



Run

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